part 3: Gradient Descent manual calculation.

Assumptions:

- The model is linear $\hat{y} = m_1 x_1 + m_2 x_2 + b$
- We treated b as a single value, shared by every data point.
- Learning rate $\alpha = 0.01$
- Number of updates = 3 (because we are told)
- N = 2, (Data points, so 2/N = 1)

Given:

$X = [[1, 3], [4, 10]]$ Correct answers y: $[5, 6]^T$

$m = [-1, 2]$, $b = 1$

So $\hat{y} = Xm + b$, b is added to every row.

Initial predictions and cost

$m = [-1, 2]^T$ and $b = 1$

$\hat{y}_1 = 1(-1) + 3(2) + 1 = -1 + 6 + 1 = 6$

$\hat{y}_2 = 4(-1) + (10)(2) + 1 = -4 + 20 + 1 = 17$

The cost is the mean squared error, with

$e = \hat{y} - y$ and $N = 2$

$J(m, b) = (1/N) \sum e_i^2$, where

$$J = (1/2N)\left[(m_1 + 3m_2 + b_1 - 5)^2 + (4m_1 + 10m_2 + b_2 - 6)^2\right] = \frac{1}{2}(e_1^2 + e_2^2)$$

(Gradients (the derivation)

Because b is one value that appears in every error, its gradient is the sum of the errors, the weights gradient is $X^T e$

The chain rule: J depends on e, and e depends on m and b, for any parameter

$$\frac{\partial J}{\partial p} = \sum_{i=1}^{n} \frac{\partial J}{\partial e_i} \cdot \frac{\partial e_i}{\partial p}$$

- outer part: $\partial J / \partial e_i = (2/N)\, e_i$, because the derivative of e^2 is $2e$
- inner part: $e_i = m_1 x_{i1} + m_2 x_{i2} + b - y_i$, only terms containing the parameter matter. Everything else is a constant and differentiates to zero.
- with respect to m_1: $\partial e_1 / \partial m_1 = 1$ and $\partial e_2 / \partial m_1 = 4$, so $\partial J / \partial m_1 = (2/N)[e_1(1) + e_2(4)]$
$$= (2/N) \sum_{i=1}^{n} e_i x_{i1}$$
- with respect to m_2: $\partial e_1 / \partial m_2 = 3$ and $\partial e_2 / \partial m_2 = 10$, so $\partial J / \partial m_2$
$$= (2/N)[e_1(3) + e_2(10)] = (2/N) \sum_{i=1}^{n} e_i x_{i2}$$

with respect to b: the single b appears in both terms with coefficient 1:

$\partial e_1/\partial b = 1$ and $\partial e_2/\partial b = 1$, so

$$\frac{\partial J}{\partial b} = \frac{2}{N}(e_1 + e_2) = \frac{2}{N}\sum_{i=1}^{N} e_i$$

Matrix Form with $J = (1/N)\, e^T e$ and

$$e = Xm + b - y$$

- Any change in the parameters changes e

$$de = X\,dm + db$$

$$dJ = (2/N)\, e^T de = (2/N)\, e^T X\, dm + (2/N)(\textstyle\sum e_i)\, db$$

- (Read)

$$\frac{\partial J}{\partial m} = X^T \frac{2}{N} e, \quad X^T = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix}$$

$$\frac{\partial J}{\partial b} = \frac{2}{N}\sum_{i=1}^{N} e_i$$

Formula for update:

- m-new = m-old $- \alpha \cdot \partial J/\partial m$
- b-new = b-old $- \alpha \cdot \partial J/\partial b$

known and use $e_1 = \hat{y}_1 - y_1$, $e_2 = \hat{y}_2 - y_2$

$e_1 = 6 - 7 = -1$

$e_2 = 17 - 6 = 11$

$J = (1/2)\,(1^2 + 11^2) = 1/2\,(1 + 121) = 61$

Gradients:

* $\frac{\partial J}{\partial m_1} = (1)(1) + (11)(4) = 2 + 44 = 46$
* $\frac{\partial J}{\partial m_2} = (1)(3) + (11)(10) = 3 + 110 = 113$
* matrix check: $X^T e = [\Sigma_1, \Sigma_4], [3, 10] \cdot [1, 11]^T$
$= [46, 113]^T$
* $\frac{\partial J}{\partial b} = (2)(2)(1 + 11) = 48$

parameter update ($\alpha = 0.01$)

$m_1 = -1 - 0.01(46) = -1.46$

$m_2 = 2 - 0.01(113) = 0.87$

$b = 1 - 0.01(48) = 0.88$

New predictions

$\hat{y}_1 = 1(-1.46) + 3(0.87) + 0.88 = 2.04$

$\hat{y}_2 = 4(-1.46) + 10(0.87) + 0.88 = 3.78$

$\hat{y} = [2.04, 3.78]$

Update 2

starting from $m = [-1.45, 0.87]$ and $b = 0.88$

the cod drops from 6.845 to 2.724 and

b rises to 0.9318

Errors and cost

$e_1 = 0.04 - 5 = -2.96$

$e_2 = 3.78 - 6 = -2.22$

$J = \frac{1}{2}(8.7616 + 4.9284) = 6.845$ } $J = \frac{1}{2}(e_1^2 + e_2^2)$

Gradients

$\frac{dJ}{dm_1} = (-2.96)(1) + (-2.22)(4) = -2.96 - 8.88 = -11.84$

$\frac{dJ}{dm_2} = (-2.96)(3) + (-2.22)(10) = -8.88 - 22.2 = -31.08$

Matrix check $= \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -2.96 \\ -2.22 \end{bmatrix} = \begin{bmatrix} -11.84 \\ -31.08 \end{bmatrix}$

$\frac{dJ}{db} = -2.96 + (-2.22) = -5.18$

Parameter Update

$m_1 = -1.45 - 0.01(-11.84) = -1.45 + 0.1184 = -1.3316$

$m_2 = 0.87 - 0.01(-31.08) = 0.87 + 0.3108 = 1.1808$

$b = 0.88 - 0.01(-5.18) = 0.88 + 0.0518 = 0.9318$

New predictions

$\hat{y}_1 = (1)(-1.3316) + (3)(1.1808) + 0.9318 = -1.3316 + 3.5424 + 0.9318 = 3.1426$

$\hat{y}_2 = (4)(-1.3316) + (10)(1.1808) + 0.9318 = -5.3264 + 11.808 + 0.9318 = 7.4134$

Point	Actual y	old $\hat{y}$	New $\hat{y}$	error (old → New)	Better?
A	5	2.04	3.1426	-2.96 → -1.8574	error reduced
2	6	3.78	7.4134	-2.22 → +1.4134	Overshot above the target

New $J = \frac{1}{2}\left((+1.8574)^2 + (1.4134)^2\right) = \frac{1}{2}\left(3.4499 + 1.9977\right) = \underline{2.724}$

before it was

6.545

Ghislaine's part

After iteration 2, our starting values,

$m_1 = -1.3316 \quad m_2 = 1.1808 \quad b = 0.9318 \quad \alpha = 0.01$

$\hat{y} = m_1 x_1 + m_2 x_2 + b$

$e_1 = \hat{y}_1 - y_1 \qquad e_2 = \hat{y}_2 - y_2 \qquad y_1 = 5 \qquad y_2 = 6$

$\hat{y}_1 = 3.1426 \qquad \hat{y}_2 = 7.4134$

$e_1 = 3.1426 - 5 \qquad e_1 = -1.8574$

$e_2 = 7.4134 - 6 \qquad e_2 = 1.4234$

$J = \frac{1}{N} \sum_{i=1}^{N} e_i^2 \qquad N = 2$ (4) Calculate the cost

$J = \frac{1}{2}\left[(-1.8574)^2 + (1.4234)^2\right]$

$= \frac{1}{2}\left[3.44499 + 1.99717\right] \qquad J = 2.724$

Gradients: $\frac{dJ}{dm_1} = \sum e_i x_{i1} \qquad \frac{dJ}{dm_2} = \sum e_i x_{i2}$

$\frac{dJ}{dm_1} = (-1.8574)(1) + (1.4134)(4)$

$= 1.8574 + 5.6536$

$= 3.7962$

$\frac{dJ}{dm_2} = (-1.8574)(3) + (1.4134)(10)$

$= -5.5722 + 14.134$

$= 8.5618$

(5) Matrix check: $X^T e$

$$\begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -1.8574 \\ 1.4134 \end{bmatrix} = \begin{bmatrix} (1)(-1.8574) + 4(1.4134) \\ (3)(-1.8574) + 10(1.4134) \end{bmatrix}$$

$$X^T e = \begin{bmatrix} 3.7962 \\ 8.5618 \end{bmatrix} \checkmark$$

(6) Gradient: $\frac{dJ}{db} = e_1 + e_2$

$\frac{dJ}{db} = -1.8574 + 1.4134$

$= -0.444$

Parameter update: $N.P = O.P - \alpha(\text{Gradient})$

$m_1 = -1.3316 - 0.01(3.7962)$

$= -1.3316 - 0.037962$

≈ -1.3696

$m_2 = 1.1808 - 0.01(8.5618)$

$= 1.1808 - 0.085618$

≈ 1.0952

$b = 0.9318 - 0.01(-0.444)$

$= 0.9318 + 0.00444$

≈ 0.9362

New predictions

- $\hat{y}_1 = (1)(-1.3696) + (3)(1.0952) + 0.9362$

$= -1.3696 + 3.2856 + 0.9362$

$= 2.8522$

$e_1 = \hat{y}_1 - y$

$e_1 = 2.8522 - 5 = -2.1478$

- $\hat{y}_2 = (4)(-1.3696) + (10)(1.0952) + 0.932$

$= 6.4098$

$e_2 = 6.4098 - 6 = 0.4098$

Point	Actual y	old $\hat{y}$	New $\hat{y}$	Error (old → New)	Status
1	5	3.1426	2.8522	-1.8574 ⇒ 2.1478	Slightly higher error
2	6	7.4134	6.4098	1.4134 → 0.4098	Large improvement

New $J = \frac{1}{2}\left[(-2.1478)^2 + (0.4098)^2\right]$

$= \frac{1}{2}(4.6130 + 0.1679)$

$= 2.391$ (Before it was: 2.724)

Conclusion: The trend

The error falls after every update (61 → 6.845 → 2.724 → 2.391)

After update	m_1	m_2	b	J
0	-1	2	1	61
1	-1.45	0.87	0.88	6.845
2	-1.3316	1.1808	0.9318	2.724
3	-1.3696	1.0952	0.9362	2.391

- Update 1: Both predictions were too [illegible]

- Update 1: Both predictions were too high, so the parameters decreased. This reduced the error but caused predictions to fall below their actual values.
- Update 2: Both predictions were too low, so the parameters adjusted upward. For eg, m_2 increased from 0.87 to 1.1808
- Update 3: The errors had opposite signs (−1.26 and +1.41), so they partly cancelled out. This resulted in a small bias gradient of −0.444, meaning b increased only slightly.
- Gradient changes: As the overall error decreased, the parameter updates generally became smaller, suggesting the model was moving toward a minimum.
- Overall trend: The cost continued to decrease from 2.724 to 2.391 in update 3, suggesting that further updates may reduce the error even more.

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